# Chapter5 Normalization and Functional Dependencies

## 1 A Single Table Approach

Imagine a university keeps all student enrollment data in a **single table** named `Enrollment`:

| StudentID | StudentName | CourseID | CourseTitle | Instructor | Grade |
|---|---|---|---|---|---|
| 101 | Ahmed | CS101 | Intro to CS | Dr. Sameer | A |
| 101 | Ahmed | MATH201 | Calculus II | Dr. Ibrahem | B+ |
| 102 | Sami | CS101 | Intro to CS | Dr. Layla | A- |
| 102 | Sami | PHY301 | Quantum Physics | Dr. Amani | C |

> **The Problem**
>
> This looks simple — but it contains serious problems known as **anomalies**.

## 2 Anomalies in Unnormalized Tables

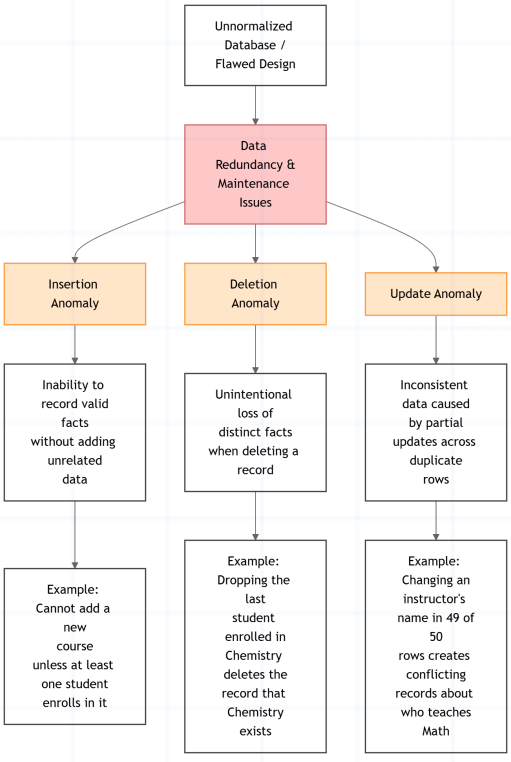



Diagram 55: Breaking down database anomalies caused by unnormalized design and data redundancy.

## 2.1 Insertion Anomaly

**Problem:** Cannot insert certain data without other unrelated data.

Suppose a new course `CS202 – Data Structures` is added, but no student has enrolled yet. You cannot insert `CourseID = CS202` and `CourseTitle = Data Structures` unless you also have a `StudentID`. The course information depends on a student being enrolled.

| StudentID | StudentName | CourseID | CourseTitle | Instructor | Grade |
|---|---|---|---|---|---|
| 101 | Ahmed | CS101 | Intro to CS | Dr. Layla | A |
| 101 | Ahmed | MATH201 | Calculus II | Dr. Ibrahem | B+ |
| 102 | Sami | CS101 | Intro to CS | Dr. Layla | A- |
| 102 | Sami | PHY301 | Quantum Physics | Dr. Amani | C |
| **Null** | **Null** | **CS202** | **Data Structures** | **Null** | **Null** |

## 2.2 Deletion Anomaly

**Problem:** Deleting data unintentionally removes other important data.

If Ahmed drops `MATH201`, and he was the only student in that course, deleting his record also deletes the course info (`MATH201`, `Calculus II`, `Dr. Ibrahem`).

| StudentID | StudentName | CourseID | CourseTitle | Instructor | Grade |
|---|---|---|---|---|---|
| 101 | Ahmed | CS101 | Intro to CS | Dr. Layla | A |
| 102 | Sami | CS101 | Intro to CS | Dr. Layla | A- |
| 102 | Sami | PHY301 | Quantum Physics | Dr. Amani | C |

## 2.3 Update Anomaly

**Problem:** Redundant data leads to inconsistencies during updates.

Dr. Layla teaches `CS101`. Her name appears in **every row** where `CS101` is offered. If Dr. Layla changes her name (e.g., to *Prof. Layla*), you must update every row with `CS101`. If you miss one row, data becomes inconsistent.

| StudentID | StudentName | CourseID | CourseTitle | Instructor | Grade |
|---|---|---|---|---|---|
| 101 | Ahmed | CS101 | Intro to CS | **Prof. Layla** | A |
| 101 | Ahmed | MATH201 | Calculus II | Dr. Ibrahem | B+ |
| 102 | Sami | CS101 | Intro to CS | **Dr. Layla** ← inconsistent! | A- |
| 102 | Sami | PHY301 | Quantum Physics | Dr. Amani | C |

## 2.4 Solution: Normalization

**Normalization** is the process of decomposing a table into smaller, well-structured tables to:

- **Eliminate redundancy** — the same data isn't stored multiple times unnecessarily
- **Avoid anomalies** — relationships between tables are clear and logical
- **Ensure data integrity** — queries run faster and are easier to maintain

We split the `Enrollment` table into three tables:

Table 27: *Students*

| StudentID | StudentName |
|---|---|
| 101 | Ahmed |
| 102 | Sami |

Table 28: *Courses*

| CourseID | CourseTitle | Instru |
|---|---|---|
| CS101 | Intro to CS | Dr. Layla |
| MATH201 | Calculus II | Dr. Ibrahem |
| PHY301 | Quantum Physics | Dr. Amani |

Table 29: *Enrollments*

| StudentID | CourseID | Grad |
|---|---|---|
| 101 | CS101 | A |
| 101 | MATH201 | B+ |
| 102 | CS101 | A- |
| 102 | PHY301 | C |

Now each anomaly disappears:

- **Adding a new course?** Just insert into `Courses` — no student needed.
- **A student drops a course?** Delete only from `Enrollments` — course data stays intact.
- **Instructor name changes?** Update one row in `Courses` — done.

> **Note**
>
> When you perform **ER-to-Relational Mapping**, following standard mapping rules automatically structures your tables to be in **at least 3NF** in the vast majority of cases:
>
> - **Entities → Tables:** Each entity type becomes a separate relation, giving entity attributes a clear home around a chosen entity key.
> - **Relationships → Foreign Keys or Junction Tables:** Binary 1:N relationships use foreign keys, and M:N relationships become separate junction tables (e.g., `Enrollments`), which naturally isolates composite key attributes like `Grade`.
> - **Attributes → Atomic Fields:** Multi-valued attributes are broken out into separate tables, satisfying 1NF.

## 3 Functional Dependencies

Before we can normalize, we need a precise language for describing what determines what in a relation. That language is **Functional Dependencies (FDs)**.

In database design, FDs provide the formal mathematical foundation used to capture real-world business rules about how attributes relate to one another.

Think of an FD like a deterministic mathematical function:

$$y = f(x) \implies \text{knowing } x \text{ always gives you exactly one } y$$

The database equivalent is written $X \to Y$ and reads:

> **"For every valid instance of $X$, there is a unique value of $Y$ in the table."**

In other words, for any two tuples (rows) $t_1$ and $t_2$ in a relation state $r(R)$:

$$\text{if } t_1[X] = t_2[X], \text{ then } t_1[Y] = t_2[Y]$$

If two rows share the same $X$ value, they **must** share the same $Y$ value—no exceptions.

> 📖 DEFINITION 1: **Functional Dependency**
>
> A Functional Dependency X→Y (read as "X functionally determines Y") holds in a relation R if, for **any** valid state of R, whenever two tuples agree on the attributes of X, they **must** also agree on the attributes of Y.
>
> - **Determinant:** The left-hand side (X).

- **Dependent:** The right-hand side (Y).

## 3.1 Why FDs Are Essential

Without functional dependencies, you cannot formally define keys or prove that a design is free of data anomalies. FDs drive the entire theoretical foundation of relational database design:

1. **Functional Dependencies ($X \to Y$):** Formally define attribute-level constraints based on domain semantics (e.g., $\text{StudentID} \to \text{StudentName}$).
2. **Keys (Super & Candidate Keys):** Derived directly from FDs by computing attribute set closures ($X^+$). A set $X$ is a superkey if $X^+ = R$, and a candidate key if $X$ is also minimal.
3. **Normal Forms ($1\text{NF} \to 2\text{NF} \to 3\text{NF} \to \text{BCNF}$):** Evaluated strictly by testing conditions on FDs and keys:
   - **2NF:** Eliminates **partial dependencies** ($\text{Non-Prime} \subset \text{Candidate Key}$).
   - **3NF:** Eliminates **transitive dependencies** ($\text{Non-Prime} \to \text{Non-Prime}$).
   - **BCNF:** Ensures that for every non-trivial FD $X \to Y$, $X$ must be a **superkey**.

📝 EXAMPLE 1: **Attribute-Level Determinism**

A company assigns each employee a unique employee number ($\text{EmpNo}$) or Social Security Number ( $\text{SSN}$).

$$\text{SSN} \to \text{Name}$$

- **Meaning:** If you know the value of $\text{SSN}$, you can uniquely identify the $\text{Name}$ of that employee.
- **Reading the Notation:** The expression $\text{SSN} \to \text{Name}$ is read as:

  > "$\text{SSN}$ *determines* $\text{Name}$" or "$\text{SSN}$ *functionally implies* $\text{Name}$."

Does the Reverse Hold ($\text{Name} \to \text{SSN}$)?

$$\text{Name} \to \text{SSN} \quad \times (\text{False})$$

**No.** It would violate database consistency if two distinct employees happened to share the same name (e.g., two employees named "Ahmed"). Because their $\text{SSN}$ values would be different, $\text{Name}$ cannot uniquely determine $\text{SSN}$.

**FDs Come from Domain Knowledge, Not Data Snapshots**

Functional dependencies are constraints on the **relation schema** itself, derived from real-world business rules—**not** from a temporary snapshot of data:

- × **Do not infer an FD** just because it happens to hold in the current rows of a table.
- ✓ **An FD is valid** only if it holds for *every possible valid state* the database could ever enter.

## 3.2 Discovering FDs from Stakeholders

Functional dependencies are not invented by database designers; they are discovered by interviewing domain experts (stakeholders) who own and understand the real-world business rules.

**EXAMPLE 2: Stakeholder Interview & FD Extraction**

By asking targeted questions to the **University Registrar**, we extract the core business constraints and translate them directly into mathematical Functional Dependencies:

| Stakeholder Question | Registrar's Business Rule | Extracted FD Notation |
|---|---|---|
| *"Can a student ID belong to multiple students?"* | *"No, each student has a unique ID and a single official name."* | $\text{StudentID} \rightarrow \text{StudentName}$ |
| *"Can a course ID have different course titles?"* | *"No, a course code like CS101 always has a single fixed title."* | $\text{CourseID} \rightarrow \text{CourseTitle}$ |
| *"Can multiple instructors teach the same course?"* | *"Yes, but in any given term, each course is taught by only one instructor."* | $(\text{CourseID}, \text{Term}) \rightarrow \text{Instructor}$ |
| *"What determines the final grade assigned to a student?"* | *"A grade is assigned to a specific student for a specific course."* | $(\text{StudentID}, \text{CourseID}) \rightarrow \text{Grade}$ |

Consider the following sample table containing student enrollment records:

| StudentID | StudentName | CourseID | CourseTitle | Term | Instructor | Grade |
|---|---|---|---|---|---|---|
| **101** | Ahmed | CS101 | Intro to CS | Fall 2026 | Dr. Layla | A |
| **101** | Ahmed | MATH201 | Calculus II | Fall 2026 | Dr. Ibrahem | B+ |
| **102** | Sami | CS101 | Intro to CS | Fall 2026 | Dr. Layla | A- |
| **102** | Sami | PHY301 | Quantum Physics | Spring 2026 | Dr. Amani | C |
| **103** | Tariq | CS101 | Intro to CS | Spring 2026 | Prof. Omar | B |

- **Data is Just an Instance:** The table rows above illustrate the rules, but the rules (FDs) are defined by domain logic, not the data snapshot.
- **Formal Validation:**
  - $\text{StudentID} \rightarrow \text{StudentName}$ holds because whenever two rows share the same $\text{StudentID}$ (e.g., $101$), they share the exact same $\text{StudentName}$ ("Ahmed").
  - $(\text{CourseID}, \text{Term}) \rightarrow \text{Instructor}$ holds because $(\text{CS101}, \text{Fall 2026})$ uniquely determines "Dr. Layla", while $(\text{CS101}, \text{Spring 2026})$ uniquely determines "Prof. Omar".

### 3.2.1 FD Validation

An Functional Dependency (FD) $X \rightarrow Y$ is **valid** (holds) in a relation state $r(R)$ if and only if for every pair of tuples (rows) $t_1$ and $t_2$ in $r(R)$ that agree on the attribute set $X$, they must also agree on the attribute set $Y$.

Formally:

$$X \rightarrow Y \iff \forall t_1, t_2 \in r(R), \quad \text{if } t_1[X] = t_2[X], \text{ then } t_1[Y] = t_2[Y]$$

**EXAMPLE 3: Validating FDs on a Relation Instance**

Consider the relation instance $R(A, B, C, D)$:

| A | B | C | D |
|---|---|---|---|
| 1 | 1 | 2 | 3 |
| 1 | 2 | 2 | 3 |
| 1 | 3 | 2 | 3 |
| 2 | 4 | 5 | 6 |
| 5 | 6 | 7 | 8 |

Which of the following FDs are satisfied by this relation?

1. $A \to B$
2. $A \to CD$
3. $AB \to CD$
4. $C \to D$
5. $B \to A$
6. $BD \to AC$
7. $AD \to BC$
8. $D \to B$
9. $D \to C$
10. $C \to A$

Show Solution

1. $A \to B$

   $\times$ Does not hold. For $A = 1$, there are three distinct values of $B$ $(1, 2, 3)$.

2. $A \to CD$

   ✓ Holds. For each value of $A$, there is a unique $(C, D)$ pair:

   - $A = 1 \implies (2, 3)$
   - $A = 2 \implies (5, 6)$
   - $A = 5 \implies (7, 8)$

3. $AB \to CD$

   ✓ Holds. Every $(A, B)$ combination is distinct across all rows, so each naturally maps to a unique $(C, D)$ pair.

4. $C \to D$

   ✓ Holds.

   - $C = 2 \implies D = 3$
   - $C = 5 \implies D = 6$
   - $C = 7 \implies D = 8$

5. $B \to A$

   ✓ Holds. Every value of $B$ is unique in this instance, mapping to a single $A$ value.

6. $BD \to AC$

   ✓ Holds. Each $(B, D)$ pair appears at most once in the table.

7. $AD \to BC$

   × Does not hold. For $(A, D) = (1, 3)$, there are three distinct $(B, C)$ pairs: $(1, 2)$, $(2, 2)$, and $(3, 2)$.

8. $D \to B$

   × Does not hold. For $D = 3$, $B$ takes values $1, 2$, and $3$.

9. $D \to C$

   ✓ Holds.

   - $D = 3 \implies C = 2$
   - $D = 6 \implies C = 5$
   - $D = 8 \implies C = 7$

10. $C \to A$

    ✓ Holds. Each $C$ value maps to a unique $A$ value ($2 \implies 1$, $5 \implies 2$, $7 \implies 5$).

> ⚠️ **Important Distinction:**
>
> While dependencies like $B \to A$ and $C \to A$ hold in this *particular sample instance*, they cannot be declared general schema FDs unless supported by true real-world business rules. A single counterexample disproves an FD, but an instance satisfying an FD does not prove it holds universally for all future data.

## 3.3 Keys, Prime, and Non-Prime Attributes

### 3.3.1 Review of Keys

> 📖 DEFINITION 2: **Key Attribute**

An attribute that uniquely identifies each entity instance.

📖 **DEFINITION 3:** **Super Key**

A set of one or more attributes (columns) that uniquely identifies a row (tuple) within a table. Any combination of columns that prevents duplicate rows is a super key.

📖 **DEFINITION 4:** **Candidate Key**

A **Candidate Key** is a *minimal* Super Key. It has the exact same unique identification property as a Super Key, but it contains **no redundant attributes**. If you remove any attribute from a Candidate Key, it ceases to be a key.

📝 **EXAMPLE 4:** **Candidate Keys vs. Super Keys**

Suppose we have a relation $R(\text{SSN}, \text{Name}, \text{Address}, \text{Phone})$ with the functional dependency:

$$\text{SSN} \rightarrow \{\text{Name}, \text{Address}, \text{Phone}\}$$

Key Classifications

1. **Candidate Key:** $\{\text{SSN}\}$
   - **Closure:** $\{\text{SSN}\}^+ = \{\text{SSN}, \text{Name}, \text{Address}, \text{Phone}\} = R$ (it uniquely identifies all attributes in the relation, so it is a **superkey**).
   - **Minimality:** No proper subset of $\{\text{SSN}\}$ can uniquely identify the entire relation. Therefore, $\{\text{SSN}\}$ is a **candidate key**.
2. **Superkey (Not a Candidate Key):** $\{\text{SSN}, \text{Name}\}$
   - **Closure:** $\{\text{SSN}, \text{Name}\}^+ = \{\text{SSN}, \text{Name}, \text{Address}, \text{Phone}\} = R$ (it uniquely identifies all attributes, so it is a **superkey**).
   - **Minimality Check:** It contains a proper subset, $\{\text{SSN}\}$, which is already a superkey on its own. Because $\text{Name}$ is redundant for uniqueness, $\{\text{SSN}, \text{Name}\}$ fails the minimality requirement and is **not a candidate key**.

## Core Concept Breakdown

$$\text{Candidate Key} = \text{Superkey} + \text{Minimality}$$

- **Superkey:** Any attribute set $X$ whose closure covers the entire relation schema ($X^+ = R$).
- **Candidate Key:** A minimal superkey—removing any attribute from $X$ causes $X^+ \neq R$.

### 3.3.2 Prime vs. Non-Prime Attributes

This distinction is critical for 2NF and 3NF normalization.

- **Prime Attribute:** An attribute that is a member of **ANY** Candidate Key.
- **Non-Prime Attribute:** An attribute that is **NOT** a member of any Candidate Key.

> 📝 EXAMPLE 5: **Candidate Keys & Attribute Classification**
>
> For a relation $R(A, B, C, D)$ with candidate keys $CK_1 = \{A, B\}$ and $CK_2 = \{B, C\}$:
>
> - **Prime Attributes:** $\{A, B, C\}$
>
>   *An attribute is prime if it is a member of **at least one** candidate key.*
>
>   - $A \in CK_1$
>   - $B \in CK_1$ and $B \in CK_2$
>   - $C \in CK_2$
> - **Non-Prime Attribute:** $\{D\}$
>
>   *An attribute is non-prime if it does **not** belong to any candidate key.*

## 3.4 Deriving Keys and Functional Dependencies

In database design, we are rarely handed all functional dependencies in their final form. We typically start with a set of known business rules $F$ and must derive implied dependencies to analyze schema properties, verify rules, and discover candidate keys.

### 3.4.1 Armstrong's Axioms (The Inference Rules)

To derive new, sound, and complete functional dependencies from a given set $F$, we use **Armstrong's Axioms**. The set of all possible FDs that can be logically inferred from $F$ is called the **closure of** $F$, denoted as $F^+$.

#### 3.4.1.1 Primary Axioms

- **Reflexivity:** If $Y \subseteq X$, then $X \to Y$.

  *(A set of attributes always determines any of its subsets.)*
- **Augmentation:** If $X \to Y$, then $XZ \to YZ$ for any attribute set $Z$.

  *(Adding the same attributes to both sides preserves the dependency.)*
- **Transitivity:** If $X \to Y$ and $Y \to Z$, then $X \to Z$.

*(Functional determination propagates across intermediate attributes.)*

📝 **EXAMPLE 6:**

Imagine a university table tracking student enrollments with columns: `StudentID`, `CourseID`, and `Semester`.

1. **Given Set ($X$):**
   - Let $X = \{\text{StudentID}, \text{CourseID}, \text{Semester}\}$
2. **Choose a Subset ($Y \subseteq X$):**
   - Let $Y = \{\text{StudentID}\}$
3. **Result ($X \to Y$):**
   - Since $Y$ is a subset of $X$, we get:

     $$\{\text{StudentID}, \text{CourseID}, \text{Semester}\} \to \{\text{StudentID}\}$$

**Intuition:** If you already know the specific combination of a student, course, and semester, you inherently know the student's ID. Knowing the whole collection of attributes guarantees you know any individual part (or combination of parts) within that collection.

📝 **EXAMPLE 7:**

1. **Given Functional Dependency ($X \to Y$):**
   - Knowing a product's barcode ($X$) determines its unit price ($Y$).
   - $\text{Barcode} \to \text{Price}$
2. **Add Attribute $Z$ to Both Sides ($XZ \to YZ$):**
   - Let $Z = \{\text{StoreLocation}\}$.
   - Augmenting both sides with the store location gives:

     $$\{\text{Barcode}, \text{StoreLocation}\} \to \{\text{Price}, \text{StoreLocation}\}$$

**Intuition:** If a barcode determines a price, then knowing the barcode *and* the store location will determine the price *and* that same store location.

📝 **EXAMPLE 8:**

Imagine a corporate database tracking employee assignments with columns: `EmployeeID`, `DeptID`, and `BuildingNo`.

1. **Given Functional Dependencies:**
   - Knowing an employee's ID determines their department ID ($X \to Y$):

     $$\text{EmployeeID} \to \text{DeptID}$$

   - Knowing the department ID determines the building where that department is located ($Y \to Z$):

     $$\text{DeptID} \to \text{BuildingNo}$$

2. **Result ($X \to Z$):**
   - By transitivity, we can chain these together to conclude that knowing the employee's ID indirectly determines their building:

     $$\text{EmployeeID} \to \text{BuildingNo}$$

**Intuition:** If $X$ leads to $Y$, and $Y$ leads to $Z$, then $X$ transitively leads to $Z$ because $Y$ acts as a bridge.

#### 3.4.1.2 Secondary (Derived) Rules

From the three primary axioms, we derive practical helper rules:

- **Union (Additive):** If $X \to Y$ and $X \to Z$, then $X \to YZ$.
- **Decomposition:** If $X \to YZ$, then $X \to Y$ and $X \to Z$.
- **Pseudotransitivity:** If $X \to Y$ and $WY \to Z$, then $WX \to Z$.

📝 EXAMPLE 9: **Union (Additive) Rule**

**Given Functional Dependencies:**

- Knowing an `EmployeeID` determines their `Name`:

  $$\text{EmployeeID} \to \text{Name}$$

- Knowing the same `EmployeeID` determines their `Email`:

  $$\text{EmployeeID} \to \text{Email}$$

- **Result ($X \to YZ$):**
  - By the Union rule, knowing the `EmployeeID` allows you to determine both together:

    $$\text{EmployeeID} \to \{\text{Name}, \text{Email}\}$$

**Intuition:** If one key can look up your name and the same key can look up your email, you can group them into a single lookup for both pieces of information.

📝 EXAMPLE 10: **Pseudotransitivity Rule**

**Given Functional Dependencies:**

- Knowing a `StudentID` ($X$) determines their current `Major` ($Y$):

  $$\text{StudentID} \rightarrow \text{Major}$$

- Knowing a `Major` ($Y$) *combined with* a specific `CourseLevel` ($W$) determines the maximum `AllowedCredits` ($Z$):

  $$\{\text{Major}, \text{CourseLevel}\} \rightarrow \text{AllowedCredits}$$

- **Result ($WX \rightarrow Z$):**
  - By Pseudotransitivity, we can substitute `Major` with `StudentID`, meaning knowing the student's ID *and* the course level gives you the allowed credits:

    $$\{\text{StudentID}, \text{CourseLevel}\} \rightarrow \text{AllowedCredits}$$

**Intuition:** Since $X$ provides $Y$, you can feed $X$ into the second rule wherever $Y$ is required, bridging $X$ and $W$ together to reach $Z$.

### 3.4.2 Practical Tips for Spotting Keys Quickly

Before running any algorithm, experienced database designers use these **visual heuristics** to immediately narrow down the search for candidate keys. These tips come from analyzing the left-hand side (LHS) and right-hand side (RHS) of the given FDs.

💡 **Tip 1: The "Must-Have" Rule (Never on RHS)**

Any attribute that **never appears on the right-hand side** of *any* FD **must** be part of **every** Candidate Key. *Why?* Because there is no way to derive this attribute from any other attribute. The only way it can appear in a closure is if you include it in the starting set.

💡 **Tip 2: The "Never a Key" Rule (Only on RHS)**

Any attribute that **only appears on the right-hand side** (and never on the left-hand side) of any FD can **never** be part of any Candidate Key. *Why?* Because this attribute is always "determined by" something else. It adds no determining power. Including it in a key would make the key non-minimal.

**Tip 3: The "Both Sides" Rule**

Attributes that appear on **both** the LHS and RHS of various FDs are the "candidate" attributes. They *might* be part of a candidate key, depending on closure computation.

**Tip 4: Single-Attribute Quick Check**

Before trying combinations, always compute the closure of each individual "Must-Have" attribute and each "Both Sides" attribute. If any single attribute's closure equals R, that attribute alone is a Candidate Key.

### 3.4.3 The Attribute Closure Algorithm

Computing the full set of all dependencies $F^+$ can be exponentially large. To determine whether a specific set of attributes $X$ can uniquely identify the entire table—or to test if a specific FD $X \to Y$ holds—we compute the **Attribute Closure** of $X$, denoted as $X^+$.

📖 DEFINITION 5: **The attribute closure**

The attribute closure $X^+$ is the set of all attributes in $R$ that are functionally determined by $X$ under the given set of dependencies $F$.

**Algorithm: Computing Attribute Closure**

1. **Initialization:** Set $X^+ = X$.
2. **Iterative Expansion:** Repeatedly search $F$ for an FD $A \to B$ such that $A \subseteq X^+$ and $B \nsubseteq X^+$. If found, set $X^+ = X^+ \cup B$.
3. **Termination:** Repeat Step 2 until a full pass over $F$ yields no new attributes (a fixpoint is reached).

📝 EXAMPLE 11: **Basic Step-by-Step Expansion**

**Given:**

Relation Schema: $R(A, B, C, D, E, F)$

Functional Dependency Set $F$:

1. $A \to B$
2. $\{B, C\} \to D$
3. $D \to E$
4. $E \to F$

Compute the attribute closure of $\{A, C\}$, denoted as $\{A, C\}^+$.

✅ Show Solution

- **Initialization:**

$$\{A, C\}^+ = \{A, C\}$$

- **Iteration 1:**
  - Check $A \to B$: LHS $A \subseteq \{A, C\}^+$. Add $B$.

$$\{A, C\}^+ = \{A, B, C\}$$

  - Check $\{B, C\} \to D$: LHS $\{B, C\} \subseteq \{A, B, C\}^+$. Add $D$.

$$\{A, C\}^+ = \{A, B, C, D\}$$

  - Check $D \to E$: LHS $D \subseteq \{A, B, C, D\}^+$. Add $E$.

$$\{A, C\}^+ = \{A, B, C, D, E\}$$

  - Check $E \to F$: LHS $E \subseteq \{A, B, C, D, E\}^+$. Add $F$.

$$\{A, C\}^+ = \{A, B, C, D, E, F\}$$

**Termination:**

$\{A, C\}^+ = \{A, B, C, D, E, F\} = R$.

> **Conclusion:** Since $\{A, C\}^+$ contains all attributes of $R$, $\{A, C\}$ is a **superkey**. Because neither $A^+ = \{A, B\}$ nor $C^+ = \{C\}$ covers $R$, $\{A, C\}$ is also a **candidate key**.

📝 EXAMPLE 12: **Partial Closure & Unused Dependencies**

**Given:**

Relation Schema: $R(A, B, C, D, E)$

Functional Dependency Set $F$:

1. $A \to B$
2. $B \to C$
3. $\{C, D\} \to E$
4. $E \to A$

Compute:

- The attribute closure of $\{A, D\}$, denoted as $\{A, D\}^+$.
- The attribute closure of $\{A, C\}$, denoted as $\{A, C\}^+$.

✅ Show Solution

**Task:** Compute the attribute closure of $\{A, D\}$, denoted as $\{A, D\}^+$.

**Trace:**

- **Initialization:**

$$\{A, D\}^+ = \{A, D\}$$

- **Iteration 1:**
  - Check $A \to B$: LHS $A \subseteq \{A, D\}^+$. Add $B$.

$$\{A, D\}^+ = \{A, B, D\}$$

  - Check $B \to C$: LHS $B \subseteq \{A, B, D\}^+$. Add $C$.

$$\{A, D\}^+ = \{A, B, C, D\}$$

  - Check $\{C, D\} \to E$: LHS $\{C, D\} \subseteq \{A, B, C, D\}^+$. Add $E$.

$$\{A, D\}^+ = \{A, B, C, D, E\}$$

- **Termination:**

$\{A, D\}^+ = \{A, B, C, D, E\} = R$.

**Task:** Compute the attribute closure of $\{A, C\}$, denoted as $\{A, C\}^+$.

**Trace:**

- **Initialization:**

$$\{A, C\}^+ = \{A, C\}$$

- **Iteration 1:**
  - Check $A \to B$: LHS $A \subseteq \{A, C\}^+$. Add $B$.

$$\{A, C\}^+ = \{A, B, C\}$$

  - Check $B \to C$: $C$ is already in the closure. No change.
  - Check $\{C, D\} \to E$: LHS $\{C, D\} \not\subseteq \{A, B, C\}^+$ (missing $D$). Cannot apply.
  - Check $E \to A$: LHS $E \not\subseteq \{A, B, C\}^+$. Cannot apply.
- **Iteration 2:**

  No new attributes can be added. Pass completes without changes.
- **Termination:**

$$\{A, C\}^+ = \{A, B, C\} \neq R$$

> **Conclusion:** $\{A, C\}$ cannot determine $D$ or $E$, so it is **not** a superkey.

### 3.4.4 The Complete Key-Finding Algorithm

This is the systematic procedure to find all Candidate Keys, classify Prime and Non-Prime attributes, and enumerate all Super Keys.

**Step 1 — Classify all attributes into three groups:**

- **Group M (Must-Have):** Attributes that never appear on the RHS of any FD. These must belong to every Candidate Key.
- **Group N (Never-a-Key):** Attributes that only appear on the RHS (never on the LHS). These can never be part of any Candidate Key.
- **Group B (Both-Sides):** Attributes that appear on both LHS and RHS across different FDs.

**Step 2 — Compute the closure of M:**

- Compute $M^+$ using the attribute closure algorithm.
- If $M^+ = R$, then $M$ is the only Candidate Key. **Stop.**
- If $M^+ \neq R$, proceed to Step 3.

**Step 3 — Find Candidate Keys by extending M:**

1. Systematically add attributes from Group B to $M$, one at a time, then in pairs, and so on.
2. For each combination, compute the attribute closure.
3. If the closure equals $R$ (i.e., $X^+ = R$), you have found a **Super Key**.
4. **Check Minimality:** Test each attribute in your Super Key by trying to remove it. If removing any attribute still results in a closure of $R$, then the attribute was redundant, meaning the set is *not* minimal and therefore **not a Candidate Key**.
5. *Never* add attributes from Group N, as they cannot help form a minimal key.

**Step 4 — Identify Prime and Non-Prime attributes:**

- **Prime Attribute:** Any attribute that appears in *at least one* Candidate Key.
- **Non-Prime Attribute:** Any attribute that appears in *no* Candidate Key.

**Step 5 — Enumerate all Super Keys (if needed):**

- Every Candidate Key is automatically a Super Key.

- Every superset of a Candidate Key is also a Super Key.
- **The Complete Set of Super Keys:** All possible unions of Candidate Keys combined with any combination of Non-Prime attributes (along with combinations with each other).

### 📝 EXAMPLE 13: Full Worked Example

Relation $R(A, B, C, D, E)$ with the set of functional dependencies:

$$F = \{A \to B,\ B \to A,\ AC \to D,\ D \to E\}$$

Find all Candidate Keys, classify Prime and Non-Prime attributes, and list all Super Keys.

✅ Show Solution

Step 1: Classify Attributes

First, analyze where each attribute appears across the left-hand side (LHS) and right-hand side (RHS) of the FDs:

| Attribute | Appears on LHS? | Appears on RHS? | Group |
|---|---|---|---|
| **A** | Yes ($A \to B$) | Yes ($B \to A$) | **Group B** (Both-Sides) |
| **B** | Yes ($B \to A$) | Yes ($A \to B$) | **Group B** (Both-Sides) |
| **C** | Yes ($AC \to D$) | **No** | **Group M** (Must-Have) |
| **D** | Yes ($D \to E$) | Yes ($AC \to D$) | **Group B** (Both-Sides) |
| **E** | **No** | Yes ($D \to E$) | **Group N** (Never-a-Key) |

**Immediate Conclusions from Heuristics:**

1. $C$ must be in every Candidate Key (it never appears on the RHS).
2. $E$ can never be in any Candidate Key (it only appears on the RHS).
3. We only need to test combinations of $C$ with attributes from Group B: $\{A, B, D\}$.

Step 2: Compute the Closure of $M = \{C\}$

| Iteration | Current X+ | FD Applied | New X+ |
|---|---|---|---|
| Start | $\{C\}$ | — | $\{C\}$ |
| 1 | $\{C\}$ | No FD has LHS $\subseteq \{C\}$ | $\{C\}$ |

- **Result:** $\{C\}^+ = \{C\} \neq R$. Since the closure does not cover the entire relation, we must extend $M$ with attributes from Group B.

Step 3: Extend $M$ and Find Candidate Keys

We systematically test adding attributes from Group B ($\{A, B, D\}$) to our core set $C$.

Try $\{A, C\}$:

| Iteration | Current X+ | FD Applied | New X+ |
|---|---|---|---|
| Start | $\{A, C\}$ | — | $\{A, C\}$ |
| 1 | $\{A, C\}$ | $A \to B$ | $\{A, B, C\}$ |
| 2 | $\{A, B, C\}$ | $AC \to D$ (since $A, C \in X^+$) | $\{A, B, C, D\}$ |
| 3 | $\{A, B, C, D\}$ | $D \to E$ | $\{A, B, C, D, E\}$ |

- **Result:** $\{A, C\}^+ = \{A, B, C, D, E\} = R$. ✅ **Super Key found!**
- **Minimality Check:**
  - Remove $A$: $\{C\}^+ = \{C\} \neq R$ ❌ (Cannot remove $A$)
  - Remove $C$: $\{A\}^+ = \{A, B\} \neq R$ ❌ (Cannot remove $C$)
- ✅ Therefore, $\{A, C\}$ **is a Candidate Key.**

Try $\{B, C\}$:

| Iteration | Current X+ | FD Applied | New X+ |
|---|---|---|---|
| Start | $\{B, C\}$ | — | $\{B, C\}$ |
| 1 | $\{B, C\}$ | $B \to A$ | $\{A, B, C\}$ |
| 2 | $\{A, B, C\}$ | $AC \to D$ | $\{A, B, C, D\}$ |
| 3 | $\{A, B, C, D\}$ | $D \to E$ | $\{A, B, C, D, E\}$ |

- **Result:** $\{B, C\}^+ = R$. ✅ **Super Key found!**
- **Minimality Check:**
  - Remove $B$: $\{C\}^+ = \{C\} \neq R$ ❌
  - Remove $C$: $\{B\}^+ = \{B, A\} \neq R$ ❌
- ✅ Therefore, $\{B, C\}$ **is a Candidate Key.**

Try $\{D, C\}$:

| Iteration | Current X+ | FD Applied | New X+ |
|---|---|---|---|
| Start | $\{D, C\}$ | — | $\{D, C\}$ |
| 1 | $\{D, C\}$ | $D \to E$ | $\{C, D, E\}$ |
| 2 | $\{C, D, E\}$ | No FD has LHS $\subseteq \{C, D, E\}$ producing $A$ or $B$ | $\{C, D, E\}$ |

- **Result:** $\{D, C\}^+ = \{C, D, E\} \neq R$. ❌ **Not a Super Key.**

*(Note: We can skip larger pairs like $\{A, D, C\}$ because they would contain the valid Candidate Key $\{A, C\}$ as a subset, making them non-minimal).*

Step 4: Classify Prime and Non-Prime Attributes

| Attribute | In CK {A,C}? | In CK {B,C}? | Classification |
|---|---|---|---|
| **A** | ✅ Yes | — | **Prime** |
| **B** | — | ✅ Yes | **Prime** |
| **C** | ✅ Yes | ✅ Yes | **Prime** |
| **D** | ❌ No | ❌ No | **Non-Prime** |
| **E** | ❌ No | ❌ No | **Non-Prime** |

- **Prime Attributes:** $\{A, B, C\}$
- **Non-Prime Attributes:** $\{D, E\}$

Step 5: Enumerate All Super Keys

Every Candidate Key is a Super Key, and every superset of a Candidate Key is also a Super Key. To find all Super Keys, we take our Candidate Keys ($\{A, C\}$ and $\{B, C\}$) and append **all possible combinations of the Non-Prime attributes** ($\{D, E\}$):

1. **Supersets of $\{A, C\}$:**
   - $\{A, C\}$ *(Candidate Key)*
   - $\{A, B, C\}$ *(Added $B$)*
   - $\{A, C, D\}$ *(Added non-prime $D$)*
   - $\{A, C, E\}$ *(Added non-prime $E$)*
   - $\{A, B, C, D\}$ *(Added $B, D$)*
   - $\{A, B, C, E\}$ *(Added $B, E$)*
   - $\{A, C, D, E\}$ *(Added $D, E$)*
   - $\{A, B, C, D, E\}$ *(Added all)*
2. **Supersets of $\{B, C\}$ (Unique additions):**
   - $\{B, C\}$ *(Candidate Key)*
   - $\{B, C, D\}$ *(Added non-prime $D$)*
   - $\{B, C, E\}$ *(Added non-prime $E$)*
   - $\{B, C, D, E\}$ *(Added $D, E$)*

- **Complete Unique Set of All Super Keys (12 total):**

$$SK = \{A, C\}, \{B, C\}, \{A, B, C\}, \{A, C, D\}, \{A, C, E\}, \{B, C, D\}, \{B, C, E\}, \{A, B, C, D\}, \{A,$$

💡 **The Pattern:** Notice that every Super Key contains either $\{A, C\}$ or $\{B, C\}$ as a subset. The Non-Prime attributes $\{D, E\}$ act as "passengers"—appending them to any Candidate Key creates a valid Super Key, but they contribute no determining power of their own.

📝 EXAMPLE 14:

Consider a relation schema $R(A, B, C, D, E)$ with the following set of functional dependencies $F$:

$$F = \{A \to BC,\ CD \to E,\ B \to D,\ E \to A\}$$

Find all candidate keys of the relation $R$.

✅ Show Solution

we need to find the attribute sets whose closure includes all attributes $\{A, B, C, D\}$ and no proper subset has the same property.

### Step 1: Identify Essential Attributes

Let's analyze which attributes appear on the **right-hand side (RHS)** versus the **left-hand side (LHS)** of the FDs:

- **RHS attributes:** $A, B, D$
- **LHS-only / Missing from RHS:** $C$ never appears on the right side of any functional dependency. Because $C$ is never generated by any FD, $C$ **must be part of every candidate key**.

### Step 2: Test Attribute Closures Including $C$

Since $C$ must be in every candidate key, we start by testing combinations of $C$ with other attributes.

1. Try $C$ alone:

- $C^+ = \{C\}$ (Does not include $A, B, D$) $\to$ Not a key.

2. Try two-attribute combinations containing $C$:

- $AC^+$:
  - Start: $\{A, C\}$
  - Apply $AC \to D$: Add $D \to \{A, C, D\}$
  - Apply $AD \to B$: Add $B \to \{A, B, C, D\}$
  - **Result:** $\{A, B, C, D\}$ (All attributes!)
  - *Check for minimality:* Since neither $A^+$ nor $C^+$ alone gives all attributes, $AC$ **is a candidate key**.
- $BC^+$:
  - Start: $\{B, C\}$
  - Apply $BC \to A$: Add $A \to \{A, B, C\}$
  - Apply $AC \to D$ (since we now have $A$ and $C$): Add $D \to \{A, B, C, D\}$
  - **Result:** $\{A, B, C, D\}$ (All attributes!)
  - *Check for minimality:* Since neither $B^+$ nor $C^+$ alone gives all attributes, $BC$ **is a candidate key**.
- $CD^+$:
  - Start: $\{C, D\}$

- Apply $CD \to B$: Add $B \to \{B, C, D\}$
- Apply $BC \to A$ (since we now have $B$ and $C$): Add $A \to \{A, B, C, D\}$
- **Result:** $\{A, B, C, D\}$ (All attributes!)
- *Check for minimality:* Since neither $C^+$ nor $D^+$ alone gives all attributes, $CD$ **is a candidate key**.

### Conclusion

The candidate keys for relation $R$ are:

1. $AC$
2. $BC$
3. $CD$

## 3.5 Types of Functional Dependencies

Not all functional dependencies are created equal. Some represent structural flaws (anomalies) in the database design, while others are mathematically trivial. Understanding these types is the exact mechanism used to define **Second Normal Form (2NF)** and **Third Normal Form (3NF)**.

*Table 30: Types of FDs*

| Term | Mathematical Definition | Meaning & Impact |
|---|---|---|
| **Trivial FD** | $X \to Y$ where $Y \subseteq X$ | Always true by definition. Carries no new information. |
| **Non-trivial FD** | $X \to Y$ where $Y \not\subseteq X$ | The meaningful FDs that define actual business rules. |
| **Full FD** | $X \to Y$ and no proper subset of $X$ determines $Y$ | The ideal state. $Y$ depends on the *whole* key. |
| **Partial FD** | $X \to Y$ and a proper subset of $X$ determines $Y$ | **Red flag for 2NF.** $Y$ depends on only *part* of a composite key. |
| **Transitive FD** | $X \to Y \to Z$ (where $Y \not\to X$) | **Red flag for 3NF.** $Z$ depends on a non-key attribute. |

### 3.5.1 Trivial Functional Dependency

📖 DEFINITION 6: **Trivial Functional Dependency**

An FD $X \rightarrow Y$ is trivial if $Y$ is a subset of $X$ ($Y \subseteq X$).

📝 EXAMPLE 15:

Consider an employee table where we know both the employee's ID and their name.

- **FD:** $\{\text{EmpID}, \text{Name}\} \rightarrow \text{EmpID}$ (Since $\text{EmpID}$ is part of the left-hand side set, this is trivial).

| EmpID | Name | Department |
|---|---|---|
| 101 | Ahmed | Computer Science |
| 102 | Mahmoud | Engineering |

If you already know that an employee has ID `101` and is named **Ahmed**, stating that $\{\text{EmpID}, \text{Name}\}$ determines $\text{EmpID}$ tells you nothing new. You already had the ID.

### 3.5.2 Non-Trivial Functional Dependency

📖 DEFINITION 7: **Non-Trivial Functional Dependency**

An FD $X \rightarrow Y$ is non-trivial if $Y$ is not a subset of $X$ ($Y \not\subseteq X$). *If $X$ and $Y$ share no common attributes at all, i.e., $X \cap Y = \emptyset$, it is called **completely non-trivial**).*

📝 EXAMPLE 16:

Consider an employee table:

- **FD:** $\text{EmpID} \rightarrow \text{Department}$ (Knowing the employee ID uniquely determines their department).

| EmpID | Name | Department |
|---|---|---|
| 101 | Ahmed | Computer Science |
| 102 | Mahmoud | Engineering |

The attribute on the right ($\text{Department}$) is completely separate from the left ($\text{EmpID}$). This is a genuine business rule used to design the database.

### 3.5.3 Full Functional Dependency

📖 DEFINITION 8: **Full Functional Dependency**

An FD $X \to Y$ is a full functional dependency if it is non-trivial, and removing *any* attribute $A$ from $X$ breaks the dependency (meaning no proper subset of $X$ determines $Y$).

📝 EXAMPLE 17:

Imagine a university enrollment relation where a student takes a course across a specific semester. The Primary Key is a composite key: $\{\text{StudentID}, \text{CourseID}\}$.

- **FD:** $\{\text{StudentID}, \text{CourseID}\} \to \text{Grade}$

| StudentID | CourseID | Grade |
|---|---|---|
| 201 (Ahmed) | CS101 | A |
| 201 (Ahmed) | MATH202 | B |
| 202 (Sara) | CS101 | A |

Can you know **Ahmed**'s Grade by looking at $\text{StudentID}$ alone? No, because Ahmed takes multiple courses with different grades. Can you know the Grade by looking at $\text{CourseID}$ alone? No, because many students take CS101 and get different grades. **Grade** requires *both* attributes together to be uniquely determined.

### 3.5.4 Partial Functional Dependency

📖 DEFINITION 9: **Partial Functional Dependency**

An FD $X \to Y$ is a partial dependency if a *proper subset* of $X$ can already determine $Y$.

📝 EXAMPLE 18:

Using the same enrollment table, suppose we also store the student's name:

- **FD:** $\{\text{StudentID}, \text{CourseID}\} \to \text{StudentName}$

| StudentID | CourseID | StudentName |
|---|---|---|
| 201 | CS101 | Ahmed |
| 201 | MATH202 | Ahmed |
| 202 | CS101 | Sara |

- **The Red Flag:** $\text{StudentName}$ depends *only* on $\text{StudentID}$, which is just a proper subset of the composite key $\{\text{StudentID}, \text{CourseID}\}$.
- **The Problem:** If **Ahmed** changes his name, you have to update it across multiple rows. If you miss one, your database becomes inconsistent. This violates **2NF**, and the fix is to move $\text{StudentName}$ into a separate table where $\text{StudentID}$ is the sole primary key.

### 3.5.5 Transitive Functional Dependency

📖 DEFINITION 10: **Transitive Functional Dependency**

An FD $X \to Z$ is transitive if there is an intermediate attribute set $Y$ such that $X \to Y$ and $Y \to Z$ (where $Y \not\to X$, and $Y$ is not a candidate key).

📝 EXAMPLE 19:

Consider an employee table where $\text{EmpID}$ is the primary key:

- **FD 1:** $\text{EmpID} \to \text{DeptID}$ (An employee belongs to a department ID)
- **FD 2:** $\text{DeptID} \to \text{DeptName}$ (A department ID determines the department name)
- **Transitive FD:** $\text{EmpID} \to \text{DeptName}$ (Through $\text{DeptID}$)

| EmpID | Name | DeptID | DeptName |
|---|---|---|---|
| 101 | Ahmed | D01 | IT |
| 102 | Khaled | D02 | HR |
| 103 | Yousef | D01 | IT |

- **The Red Flag:** $\text{DeptName}$ is a non-prime attribute, but it depends on $\text{DeptID}$, which is *also* a non-prime attribute (just a regular column, not a candidate key for this table).
- **The Problem:** If Department `D01` changes its name from **IT** to **Computer Science**, you have to update every single employee belonging to that department. This violates **3NF**, and the fix is to extract department data into its own table ($\text{DEPARTMENT}$).

# 4 Normalization

**Normalization** is the systematic approach of effectively designing a database so that it minimizes data redundancy and inconsistency. It is generally done through a series of steps called **Normal Forms (NF)**.

Each normal form builds on the previous one—a relation in a higher NF automatically satisfies all lower NFs:

$$1NF \subset 2NF \subset 3NF \subset BCNF \subset 4NF \subset 5NF$$

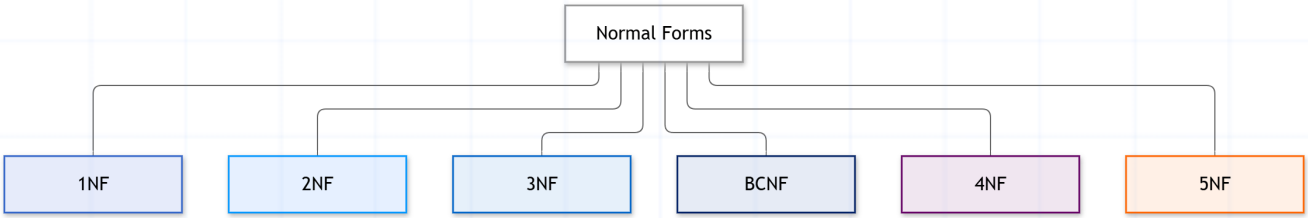



Diagram 56: Hierarchy of Database Normal Forms

| Normal Form | Key Requirement | What it Eliminates |
|---|---|---|
| **1NF** | All attribute values are atomic | Multi-valued / repeating groups |
| **2NF** | 1NF + no partial FDs on any candidate key | Redundancy from partial dependencies |
| **3NF** | 2NF + no transitive FDs (with prime-attr exception) | Transitive dependencies via non-key attributes |
| **BCNF** | Every FD determinant is a superkey | All non-trivial FD-based redundancy |
| **4NF** | BCNF + no non-trivial MVDs | Multi-valued dependency redundancy |
| **5NF** | 4NF + no non-trivial join dependencies | Decomposition artifacts |

## 4.1 The Origins of Database Tables

Before applying normalization rules, it is essential to understand where database tables originate. Tables in a normalized relational system typically come from one of two design paths:

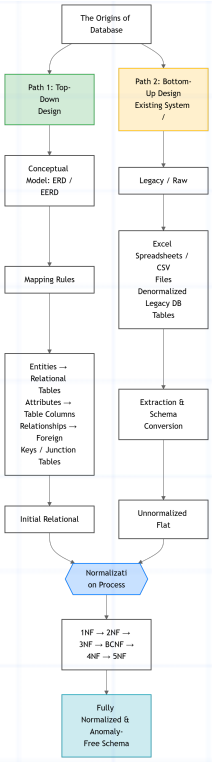



Diagram 57: Top-Down (ERD) vs. Bottom-Up (Legacy/Flat files) database design approaches leading into the normalization process (1NF through 5NF).

## 4.2 First Normal Form (1NF)

A relation is in 1NF if:

- Every attribute contains only **atomic** (single-valued, indivisible) values
- There are no repeating groups or multi-valued attributes
- There is a **primary key** that uniquely identifies each row

### 📝 EXAMPLE 20: Violating 1NF

Consider an unnormalized student table where a student can enroll in multiple courses, and those courses are stored together inside a single comma-separated cell:

| StudentNo | StudentName | Courses |
|---|---|---|
| S1 | Ahmed | DS, Algorithms |
| S2 | Hussein | Databases |

✅ Show Solution

**The Issue:** The `Courses` attribute is **not atomic** because it holds multiple values in a single cell. This directly violates 1NF and prevents standard relational querying (like indexing or joining on individual course identifiers).

To bring the relation into 1NF, we split the multi-valued field into separate rows so that each cell holds exactly one atomic value:

| StudentNo | StudentName | CourseNo | CourseName |
|---|---|---|---|
| S1 | Ahmed | CS101 | Data Structures |
| S1 | Ahmed | CS102 | Algorithms |
| S2 | Hussein | CS301 | Databases |

- **Atomicity Achieved:** Each cell now holds a single, indivisible value.
- **Primary Key:** The composite key $\{StudentNo, CourseNo\}$ uniquely identifies each row in the flattened relation.

> ⓘ **Note on Redundancy**
>
> While this table is now successfully in 1NF, the `StudentName` attribute is repeated for student `S1`. This structural redundancy is systematically addressed by moving to **Second Normal Form (2NF)**.

## 4.3 Second Normal Form (2NF)

A relation schema $R$ is in Second Normal Form (2NF) if and only if:

- It is already in **1NF**.
- Every non-prime attribute is **fully functionally dependent** on every candidate key.

> ⓘ **Understanding Partial Dependencies**
>
> - **Scope:** 2NF is only relevant when a relation has a **composite primary key** (composed of two or more attributes). A relation with a single-attribute primary key is automatically in 2NF if it is already in 1NF.
> - **Partial Dependency:** This occurs when a non-prime (non-key) attribute is functionally dependent on only a *proper subset* of a composite candidate key, rather than the entire key.
>
> Second Normal Form eliminates these partial dependencies by breaking them out into separate relations.

📝 EXAMPLE 21: **Violating 2NF**

Consider the 1NF student enrollment table where the composite key is $StudentNo, CourseNo$:

| StudentNo | StudentName | CourseNo | CourseName | InstructorNo | Grade |
|---|---|---|---|---|---|
| S1 | Ahmed | CS101 | Data Structures | I1 | A |
| S1 | Ahmed | CS102 | Algorithms | I2 | B+ |
| S2 | Hussein | CS301 | Databases | I3 | A- |

**Functional Dependencies Present:**

- $StudentNo, CourseNo \rightarrow Grade$ (Full Dependency: both attributes are required to determine the grade).
- $StudentNo \rightarrow StudentName$ (**Partial Dependency**: `StudentName` depends only on `StudentNo`, not on `CourseNo`).
- $CourseNo \rightarrow CourseName, InstructorNo$ (**Partial Dependency**: course attributes depend only on `CourseNo`, not on `StudentNo`).

✅ Show Solution

To achieve 2NF, we decompose the table by extracting the attributes involved in partial dependencies alongside the respective subset of the key:

1. **Students Table** (Key: $StudentNo$)
   - Stores attributes dependent solely on the student identifier.

| StudentNo | StudentName |
|---|---|
| S1 | Ahmed |
| S2 | Hussein |

2. **Courses Table** (Key: $CourseNo$)

- Stores attributes dependent solely on the course identifier.

| CourseNo | CourseName | InstructorNo |
|---|---|---|
| CS101 | Data Structures | I1 |
| CS102 | Algorithms | I2 |
| CS301 | Databases | I3 |

3. **Enrollments Table** (Key: $StudentNo, CourseNo$)

- Stores the relationship and attributes fully dependent on the entire composite key.

| StudentNo | CourseNo | Grade |
|---|---|---|
| S1 | CS101 | A |
| S1 | CS102 | B+ |
| S2 | CS301 | A- |

**Result:** All non-key attributes now depend on the *entire* primary key of their respective relations. Partial dependencies are eliminated.

## 4.4 Third Normal Form (3NF)

A relation schema $R$ is in **Third Normal Form (3NF)** if and only if it is already in **2NF** and, for every non-trivial functional dependency $X \to A$, at least one of the following conditions holds:

- $X$ is a **superkey** of $R$.
- $A$ is a **prime attribute** (i.e., $A$ is a member of some candidate key of $R$).

**Core Rules & Key Concepts**

- **Eliminates Transitive Dependencies:** A transitive dependency occurs when a non-key attribute determines another non-key attribute, forming a chain like $Key \to Y \to Z$ (where $Y$ is not a superkey and $Z$ is a non-prime attribute). 3NF systematically removes these structures to prevent update anomalies.
- **The Prime Attribute Exception:** Unlike Boyce-Codd Normal Form (BCNF), 3NF permits a non-key attribute to appear on the right-hand side of a dependency as long as that attribute belongs to a candidate key. This specific rule allows 3NF to preserve functional dependencies during decomposition where BCNF might otherwise fail.

EXAMPLE 22: **Violating 3NF**

Suppose we have a `Courses` relation schema that extends our 2NF design by also embedding instructor details:

$$Courses(CourseNo, CourseName, InstructorNo, InstructorName, InstructorLocation)$$

- **Candidate Key:** $\{CourseNo\}$
- **Functional Dependencies:**
  1. $CourseNo \to \{CourseName, InstructorNo\}$ (Key determinant)
  2. $InstructorNo \to \{InstructorName, InstructorLocation\}$ (Transitive dependency)

Why It Violates 3NF

- The functional dependency $InstructorNo \rightarrow \{InstructorName, InstructorLocation\}$ features a determinant ($InstructorNo$) that is **not a superkey** of the relation.
- Furthermore, the attributes on the right-hand side are **non-prime attributes** (they do not belong to any candidate key).
- This creates a transitive chain: $CourseNo \rightarrow InstructorNo \rightarrow \{InstructorName, InstructorLocation\}$.

✅ Show Solution

To bring the relation into 3NF, we decompose it by extracting the transitive dependency into its own relation:

1. **Courses Table** (Key: $CourseNo$)
   - Retains core course information and references the instructor via a foreign key.

| CourseNo | CourseName | InstructorNo |
|---|---|---|
| CS101 | Data Structures | I1 |
| CS102 | Algorithms | I2 |
| CS301 | Databases | I1 |

2. **Instructors Table** (Key: $InstructorNo$)

- Isolates attributes that depend directly on the instructor identifier.

| InstructorNo | InstructorName | InstructorLocation |
|---|---|---|
| I1 | Dr. Ali | Building A |
| I2 | Dr. Sara | Building B |

**Result:** Transitive dependencies are successfully eliminated. Updating an instructor's location now requires modifying a single row in the `Instructors` table, preventing data inconsistency.

## 4.5 Solved Examples

📝 EXAMPLE 23:

Consider a relation schema $R(A, B, C, D, E, F)$ with the following set of functional dependencies $F$:

$$F = \{AB \rightarrow C,\ D \rightarrow E,\ B \rightarrow F,\ A \rightarrow D\}$$

1. **Find all candidate keys** of the relation $R$.

2. **Determine the highest normal form** that relation $R$ satisfies (1NF, 2NF, 3NF, or BCNF). Justify your answer.

3. If the relation is not in 3NF,**decompose it** into 3NF relations.

✅ Show Solution

To determine whether the given relation $R(A, B, C, D)$ is in Second Normal Form (2NF), let's first find the candidate keys, identify any partial dependencies, and then perform the decomposition if necessary.

### Step 1: Find the Candidate Keys of $R$

Given the functional dependencies $FD = \{AB \to CD,\ B \to C\}$:

1. **Test attribute closures:**
   - Let's check $AB^+$:
     - Start:$\{A, B\}$
     - Apply $AB \to CD$: Add $C, D \to \{A, B, C, D\}$
     - **Result:**$\{A, B, C, D\}$. Thus,$AB$ **is a candidate key**.
   - Let's check $AB$ subsets ($A^+$ and $B^+$):
     - $A^+ = \{A\}$ (cannot determine $B, C, D$)
     - $B^+ = \{B, C\}$ (using $B \to C$)

Since no proper subset of $AB$ can determine all attributes,$AB$ **is the only candidate key** for relation $R$.

- **Prime attributes (part of a candidate key):**$A, B$
- **Non-prime attributes:**$C, D$

### Step 2: Check for 2NF

**Definition of 2NF:** A relation is in 2NF if it is in 1NF and **no non-prime attribute is fully functionally dependent on a proper subset of any candidate key** (i.e., no partial dependencies).

Let's check the given FDs against this rule:

1. $AB \to CD$: This is a full dependency since the entire key $AB$ determines $CD$.
2. $B \to C$:
   - Here,$B$ is a **proper subset** of the candidate key $AB$ (i.e.,$B \subset AB$).
   - $C$ is a **non-prime attribute**.
   - Therefore,$B \to C$ is a **partial dependency** (a non-prime attribute is determined by part of the candidate key).

Conclusion:**Relation $R$ is NOT in 2NF** because of the partial dependency $B \to C$.

### Step 3: Convert $R$ into 2NF

To eliminate partial dependencies and convert the relation into 2NF, we decompose $R$ into smaller relations:

1. **Extract the partial dependency into its own relation:**
   - The determinant $B$ and the attribute it determines ($C$) form the first decomposed relation:
     - $R_1(B, C)$ with FD:$B \to C$ (where $B$ is the primary key of $R_1$).
2. **Create the remaining relation with the candidate key and remaining attributes:**
   - The original candidate key attributes plus any attributes not involved in the partial dependency form the second relation:
     - $R_2(A, B, D)$ with FDs:$AB \to D$ (where $AB$ is the primary key of $R_2$).

### Final Result

The relation $R$ is decomposed into the following 2NF relations:

- $R_1(B, C)$ with primary key $B$
- $R_2(A, B, D)$ with primary key $AB$

📝 **EXAMPLE 24:**

Consider a relation schema $R(A, B, C, D, E)$ with the following set of functional dependencies $F$:

$$F = \{AB \to C,\ C \to D,\ D \to B\}$$

1. **Find all candidate keys** of the relation $R$.
2. **Determine whether $R$ is in 3NF**. If not, decompose it into 3NF.

✅ Show Solution

**Step 1: Find the Candidate Keys**

1. Test attribute closures:
   - $AB^+$:
     - Start:$\{A, B\}$
     - Apply $AB \to C$: Add $C \to \{A, B, C\}$
     - Apply $C \to D$: Add $D \to \{A, B, C, D\}$
     - Apply $D \to B$: Add $B$ (already there)
     - *Wait, what about $E$?* Notice that $E$ **never appears on the right-hand side** of any functional dependency in $F$. Because $E$ cannot be generated by any FD,$E$ **must be part of every candidate key**.
   - Let's re-test $ABE^+$:
     - Start:$\{A, B, E\}$

- Apply $AB \to C \to D \to B \to C \ldots \to$ Generates $\{A, B, C, D, E\}$.
- **Result:** $\{A, B, C, D, E\}$. Thus, $ABE$ **is a candidate key**.

2. Are there other candidate keys?

- Since $D \to B$ and $C \to D$ and $AB \to C$, we can substitute:
- Let's check $ACE^+$:
  - Start: $\{A, C, E\}$
  - Apply $C \to D$: Add $D \to \{A, C, D, E\}$
  - Apply $D \to B$: Add $B \to \{A, B, C, D, E\}$
  - **Result:** $\{A, B, C, D, E\}$. Thus, $ACE$ **is also a candidate key**.
- Let's check $ADE^+$:
  - Start: $\{A, D, E\}$
  - Apply $D \to B$: Add $B \to \{A, B, D, E\}$
  - Apply $AB \to C$: Add $C \to \{A, B, C, D, E\}$
  - **Result:** $\{A, B, C, D, E\}$. Thus, $ADE$ **is also a candidate key**.

**Candidate Keys:** $ABE$, $ACE$, and $ADE$.

- **Prime attributes:** $A, B, C, D, E$ *(Every attribute is part of at least one candidate key!)*
- **Non-prime attributes:** None.

**Step 2: Check for 3NF**

**Definition of 3NF:** A relation is in 3NF if, for every functional dependency $X \to Y$, at least one of the following holds:

1. $X \to Y$ is a trivial functional dependency ($Y \subseteq X$).
2. $X$ is a **superkey** for $R$.
3. Every attribute in $Y - X$ is a **prime attribute** (part of a candidate key).

Let's test each FD in $F$:

1. $AB \to C$: $AB$ is not a superkey, but $C$ **is a prime attribute** $\to$ Satisfies 3NF.
2. $C \to D$: $C$ **is NOT a superkey**, and $D$ **is a prime attribute** $\to$ Satisfies 3NF! *(This is the crucial edge case: even though $C$ is not a superkey, $3NF$ allows this dependency because $D$ is prime).*
3. $D \to B$: $D$ **is NOT a superkey**, and $B$ **is a prime attribute** $\to$ Satisfies 3NF.

**Conclusion:** Relation $R$ **is already in 3NF** because every non-trivial dependency has a right-hand side consisting of prime attributes. No decomposition is needed.

📝 **EXAMPLE 25:**

Consider a relation schema $R(W, X, Y, Z)$ with the following set of functional dependencies $F$:

$$F = WX \to Y,\ Y \to Z,\ Z \to X$$

1. **Find all candidate keys** of the relation $R$.
2. **Determine the highest normal form** that relation $R$ satisfies (1NF, 2NF, 3NF, or BCNF). Justify your answer.

✅ Show Solution

**Step 1: Find the Candidate Keys**

1. Analyze the attributes to see which ones never appear on the right-hand side (RHS) of any FD:
   - RHS attributes:$Y, Z, X$.
   - Notice that $W$ **never appears on the right-hand side** of any functional dependency. Therefore,$W$ **must be part of every candidate key**.
2. Test attribute closures starting with $W$:
   - $W^+$**:**
     - Start:$\{W\}$
     - No FD has $W$ alone on the LHS. Thus,$W^+ = \{W\}$ (Not a key).
   - $WX^+$**:**
     - Start:$\{W, X\}$
     - Apply $WX \to Y$: Add $Y \to \{W, X, Y\}$
     - Apply $Y \to Z$: Add $Z \to \{W, X, Y, Z\}$
     - **Result:**$\{W, X, Y, Z\}$ (All attributes!) $\to WX$ **is a candidate key**.
   - **Alternative path using** $Z \to X$**:**
     - What about $WZ^+$?
     - Start:$\{W, Z\}$
     - Apply $Z \to X$: Add $X \to \{W, X, Z\}$
     - Apply $WX \to Y$: Add $Y \to \{W, X, Y, Z\}$
     - **Result:**$\{W, X, Y, Z\}$ (All attributes!) $\to WZ$ **is also a candidate key**.

**Candidate Keys:**$WX$ and $WZ$.

- **Prime attributes:**$W, X, Z$ (attributes that belong to at least one candidate key). Wait, let's check $Y$:
  - Does $Y$ belong to a candidate key? Let's check $WY^+$:
    - Start:$\{W, Y\}$
    - Apply $Y \to Z$: Add $Z \to \{W, Y, Z\}$
    - Apply $Z \to X$: Add $X \to \{W, X, Y, Z\}$
    - **Result:**$\{W, X, Y, Z\} \to WY$ **is also a candidate key**!

- **All attributes ($W, X, Y, Z$) are prime attributes** because every single one of them participates in at least one candidate key ($WX$,$WZ$, or $WY$).

**Step 2: Determine the Highest Normal Form**

Let's check each FD against the normalization rules:

1. $WX \rightarrow Y$:
   - $WX$ is a candidate key (superkey) $\rightarrow$ Satisfies BCNF, 3NF, and 2NF.
2. $Y \rightarrow Z$:
   - $Y$ is **not a superkey**.
   - However, let's check the right-hand side ($Z$):$Z$ **is a prime attribute** (part of candidate key $WZ$).
   - Under the 3NF definition, a dependency where the LHS is not a superkey is still **allowed in 3NF** as long as the RHS attributes are prime. Therefore,$Y \rightarrow Z$ **satisfies 3NF**, but violates BCNF.
3. $Z \rightarrow X$:
   - $Z$ is **not a superkey**.
   - However, the right-hand side ($X$) is **also a prime attribute** (part of candidate keys $WX$ and $WZ$).
   - Therefore,$Z \rightarrow X$ **satisfies 3NF**, but violates BCNF.

**Conclusion:**

- The relation **is in 3NF** because every non-trivial dependency has a right-hand side consisting of prime attributes.